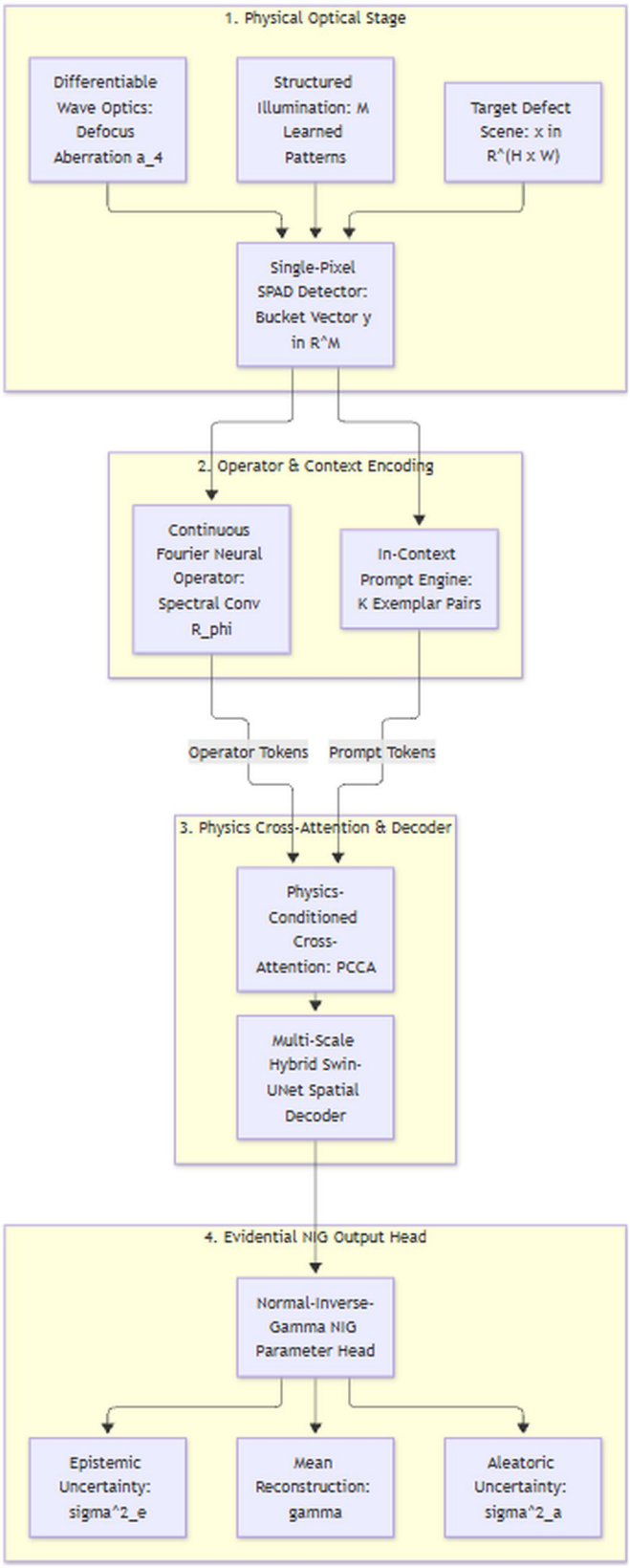

1. Physical Optical Stage
Differentiable Wave Optics: Defocus Aberration a_4
Structured Illumination: M Learned Patterns
Target Defect Scene: x in R^(H x W)
Single-Pixel SPAD Detector: Bucket Vector y in R^M
2. Operator & Context Encoding
Continuous Fourier Neural Operator: Spectral Conv R_phi
In-Context Prompt Engine: K Exemplar Pairs
Operator Tokens
Prompt Tokens
3. Physics Cross-Attention & Decoder
Physics-Conditioned Cross-Attention: PCCA
Multi-Scale Hybrid Swin-UNet Spatial Decoder
4. Evidential NIG Output Head
Normal-Inverse-Gamma NIG Parameter Head
Epistemic Uncertainty: sigma^2_e
Mean Reconstruction: gamma
Aleatoric Uncertainty: sigma^2_a